FoodHub Low Code Project

Methodology & Workflow

Foundations of AI — MIT Applied AI and Data Science Program

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Table of Contents

- 1. Executive Summary
- 2. Business Problem Overview
- 3. Solution Approach
- 4. Data Overview 4.1 Dataset Structure 4.2 Categorical Variables 4.3 Numerical Variables 4.4 Data Types 4.5 Missing Values 4.6 Summary Descriptive Statistics 4.7 Rating Distribution
- 5. Exploratory Data Analysis (EDA) 5.1 Univariate Analysis 5.2 Key Univariate Findings (Business Application)
- 6. Key Insights and Conclusions
- 7. Recommendations
- 8. Tools and Technologies

1. Executive Summary

The FoodHub food aggregator, located in New York City, operates in a densely populated area that offers key advantages, including a wide range of restaurants, high customer demand, and substantial weekend order volume. The FoodHub company comprises a group of high-performing restaurants. Data analysis of order costs, customer rating behavior, and business processed data reveals opportunities to boost customer satisfaction, streamline operations, and amplify revenue through focused marketing, advanced logistics, and bolstered restaurant partnerships.

Key Conclusions

- Strong weekend demand, driven by American cuisine among its five top-performing restaurants
- FoodHub should focus on developing strong partnerships with top-performing restaurants (e.g., Shake Shack, The Meatball Shop)
- Food delivery times are efficient, with orders averaging less than 30 minutes
- Many ratings are missing, causing inaccurate customer satisfaction analysis
- Strong baseline of positive experiences, driven by abundance of 5-star ratings
- Current total net revenue from the dataset is about $6,166.3, where most orders do not surpass $20

Key Actionable Insights

- 20 top-performing restaurants responsible for high order count and revenue
- Weekend delivery times (22 min) are faster than weekdays (28 min)
- 89.46% of orders delivered within 60 minutes
- Most popular cuisines: American, Japanese, and Italian
- Customer ratings drop when delivery times exceed ~25 minutes

- Missing ratings affect credibility of customer satisfaction metrics

Key Recommendations

- Launch loyalty program and introduce upsell bundling
- Promote top-performing restaurants through marketing campaigns
- Implement in-app prompts and push notifications to reduce “Not given” ratings
- Expand kitchen staffing during weekdays to minimize food preparation delays
- Implement optimized navigation algorithm to lower delivery times
- Create premium offers and subscription plans
- Feature combo deals in $12–$15 price range

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2. Business Problem Overview

FoodHub connects customers to restaurants across NYC via a smartphone app. Despite growing demand among students and busy professionals, the company faces the following challenges:

- **Customer retention and engagement challenges** — many customers fail to return and do not leave ratings
- **Delivery time inconsistencies** — about 10% of orders exceed 60 minutes
- **High commission fees** — causing disapproval among ~30% of partnering restaurants
- **Revenue optimization needs** — fewer than one-third of orders exceed $20; a few top-performing restaurants account for most revenue

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3. Solution Approach

FoodHub follows a data-driven, multi-faceted methodology using Python-driven statistical modeling and computations conducted by the Data Science team to address critical problems and reveal key insights, operational findings, strategies, and recommendations from the provided dataset.

The approach emphasizes Python-powered statistical analyses with code snippets and outputs provided in the report body and appendix. The methodology delivers comprehensive data analysis and insights to support strategic interventions and recommendations.

Three Core Analytical Dimensions

1. **Customer Clustering Behavioral Metrics** — identifying top customers and cuisine preferences
2. **Restaurant Performance Benchmarking** — comparing total number of orders, revenue, and ratings across restaurants
3. **Operational Metrics** — tracking food preparation and delivery times

The overall methodological approach focuses on delivering projections that address inferences on customer retention, delivery efficiency, restaurant partnering satisfaction, and revenue optimization.

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4. Data Overview

4.1 Dataset Structure

- 1,898 unique orders (rows), 9 columns (variables)
- Complete dataset with no missing null values across all columns

4.2 Categorical Variables

- **Order ID** — unique identifier for each order
- **Customer ID** — ID of the customer placing an order
- **Restaurant Name** — name of the restaurant fulfilling the order
- **Cuisine Type** — type of cuisine ordered
- **Day of the Week** — Weekend or Weekday

4.3 Numerical Variables

- **Cost of the Order** — cost per order (float64)
- **Rating** — customer rating (originally object, converted to integer); values: “Not given”, 3, 4, 5
- **Food Preparation Time** — time for restaurant to prepare food (int64, minutes)
- **Delivery Time** — time for delivery employee to deliver food (int64, minutes)

4.4 Data Types

Column	Data Type	Notes
order_id	int64	Later converted to object (identifier)
customer_id	int64	Later converted to object (identifier)
restaurant_name	object	Categorical
cuisine_type	object	Categorical
cost_of_the_order	float64	Numerical
day_of_the_week	object	Categorical (Weekend/Weekday)
rating	object	Later converted to integer for calculations
food_preparation_time	int64	Numerical (minutes)
delivery_time	int64	Numerical (minutes)

4.5 Missing Values

All columns show **0 missing values** — confirming FoodHub successfully compiled a complete dataset. This ensures cleaner analysis with higher reliability.

4.6 Summary Descriptive Statistics

Statistic	cost_of_the_order	food_preparation_time	delivery_time
Count	1,898	1,898	1,898
Mean	$16.50	27.37 min	24.16 min
Std Dev	$7.48	4.63 min	4.97 min
Min	$4.47	20 min	15 min
25%	$12.08	23 min	20 min
50% (Median)	$14.14	27 min	25 min
75%	$22.30	31 min	28 min
Max	$35.41	35 min	33 min

4.7 Rating Distribution

Rating	Count
Not given	736
5	588
4	386
3	188

Observation

736 orders without ratings; among rated orders, 5-star ratings dominate, indicating a positive satisfaction baseline. No ratings of 1 or 2 exist in the dataset.

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5. Exploratory Data Analysis (EDA)

5.1 Univariate Analysis

5.1.1 Cuisine Type Distribution

- 14 distinct cuisine types; 178 unique restaurants
- American most common (~580 orders), followed by Japanese (~470) and Italian (~300)
- Distribution highly skewed — a few cuisines dominate while many receive few orders
- Niche cuisines: Vietnamese, Korean (~20 orders each)

5.1.2 Cost of the Order

- Histogram: slightly right-skewed (positively skewed); peak density concentrated around $10–$15
- Secondary peaks around $25 and $30
- Boxplot: median ~$15, Q1 ~$12, Q3 ~$22; minimum ~$5, maximum ~$35
- Opportunities to upsell, bundle, and run premium promotions

5.1.3 Day of the Week

- Weekend dominance: ~1,400 orders (about double weekday orders)
- Dataset heavily skewed toward weekend activity

- Suggests leisure time, social gatherings, or dining out patterns
- Implication: more staffing, resources, and marketing should target weekdays

5.1.4 Rating

- “Not given” dominates (~736); rating of 5 second most common (~588)
- Rating of 4: ~386; Rating of 3: ~188
- Positive bias — ratings of 5 and 4 account for substantial share
- Evidence for incentives to encourage customers to leave ratings

5.1.5 Food Preparation Time

- Multimodal distribution with peaks at ~20, 26, 30, and 34 minutes
- Relatively symmetric, slightly right-skewed
- Boxplot: median ~27 min, Q1 ~23 min, Q3 ~31 min
- Suggests heterogeneous preparation times across cuisine types

5.1.6 Delivery Time

- Slightly right-skewed with peaks at ~25 and ~28 minutes
- Secondary peaks at ~15 and ~20 minutes
- Boxplot: median ~25 min, Q1 ~20 min, Q3 ~28 min
- Wide spread suggests high variability in service speed
- Need for route optimization, hotspot delivery, and customer communication

5.2 Key Univariate Findings (Business Application)

5.2.1 Top 5 Restaurants by Order Volume

Rank	Restaurant	Orders
1	Shake Shack	219
2	The Meatball Shop	132
3	Blue Ribbon Sushi	119
4	Blue Ribbon Fried Chicken	96
5	Parm	68

Note

Brand relationship between Blue Ribbon Sushi (Japanese) and Blue Ribbon Fried Chicken (American) — both under the Blue Ribbon brand, spanning two of the top three cuisine categories.

5.2.2 Most Popular Weekend Cuisine

American cuisine dominates weekends with **415 orders** — demonstrating strong customer inclination for American-style restaurants during leisure periods.

5.2.3 Orders Above $20

- 555 orders cost more than $20 (**29.24%** of total)
- 70.76% of orders cost $20 or less
- About 1 in 3 orders are high-value (above $20)
- Marketing strategies needed for upselling and bundling

5.2.4 Mean Delivery Time

Mean delivery time across all orders: **24.16 minutes**

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6. Key Insights and Conclusions

Cuisine and Rating Insights

- High customer demand for American cuisine on weekends (400+ orders)
- Most customers order on weekends rather than weekdays — suggesting preference for leisure dining and social gatherings
- Large number of ratings missing; ratings of 5 dominate over 3 and 4
- Most customers satisfied, but some dissatisfied due to longer total delivery times
- Top-rated restaurants (The Meatball Shop, Blue Ribbon Fried Chicken, Shake Shack) show strong reputations and customer loyalty

Operational Insights

- Average delivery: 28 minutes weekdays, 22 minutes weekends
- ~1 in 10 orders takes longer than 60 minutes
- Food preparation time ranges from 24 to 31 minutes across cuisine types
- Weekday gridlocks represent a significant improvement zone
- Need to establish baselines for food preparation times

Revenue Insights

- Top-rated restaurants (Shake Shack, The Meatball Shop) are sales generators
- Mid-tier restaurants show potential for upselling and advertising support
- Top frequent high-value customers suggest loyalty program incorporation and serve as predictive demand indicators

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7. Recommendations

4. **Promote American Cuisine on Weekends** — focus marketing efforts to increase revenue during peak demand periods
5. **Premium Advertising for Higher-Cost Cuisines** — feature Japanese and Italian restaurants in premium campaigns; position Vietnamese and Korean as affordable fast-prep alternatives
6. **Incentivize Customer Ratings** — offer discounts and in-app notifications to reduce "Not given" ratings
7. **Optimize Weekday Staffing** — ensure sufficient restaurant employees and delivery drivers to lower weekday delivery times
8. **Establish Restaurant Partnership Programs** — partner with top-rated restaurants (Shake Shack, The Meatball Shop) using loyalty programs and premium product options
9. **Introduce Bundled Meals and Combo Pricing** — target lower- to mid-tier restaurants with combo deals in the $12–$15 range
10. **Customer Loyalty Outreach** — offer top frequent customers exclusive offers and early access to new programs through subscription services
11. **Strategic Pricing** — incorporate premium food pricing options aligned with high-paying customer expectations
12. **Launch Loyalty Programs and Upsell Bundling** — attract more customers and increase average order value
13. **Implement Optimized Navigation Algorithm** — lower delivery times, especially for 60+ minute deliveries
14. **Expand Kitchen Staffing on Weekdays** — minimize food preparation delays during high-traffic weekday periods
15. **Create Premium Offers and Subscription Plans** — partner with high-performing restaurants to optimize revenue

8. Tools and Technologies

Category	Tool / Technology
Programming Language	Python (pandas, matplotlib, seaborn)
Development Environment	Google Colab Notebook
Analytical Techniques	Descriptive statistics and EDA techniques
Data Visualization	Histograms, boxplots, bar charts